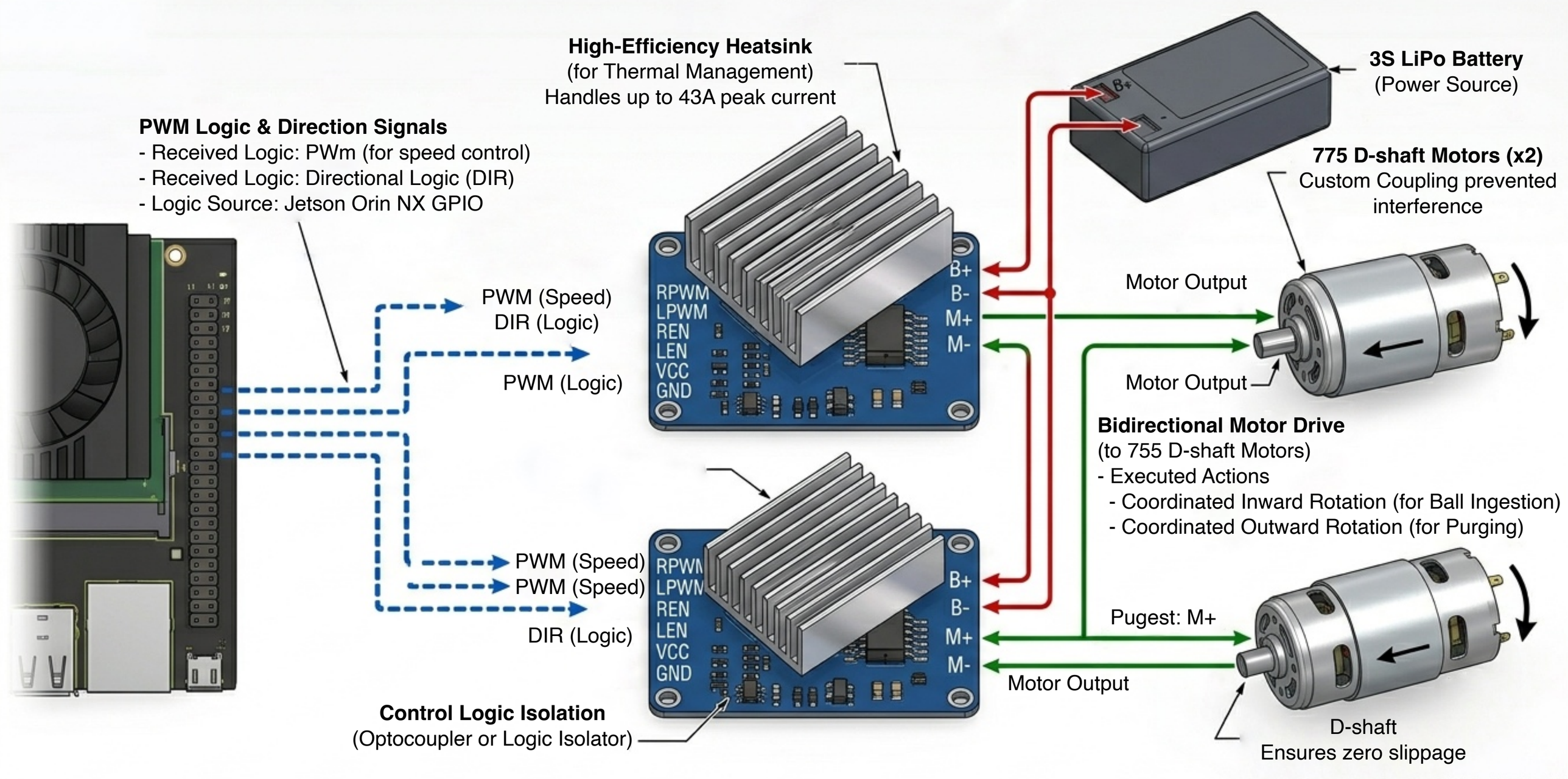
High-Efficiency Heatsink
(for Thermal Management)
Handles up to 43A peak current
3S LiPo Battery
(Power Source)
PWM Logic & Direction Signals
- Received Logic: PWm (for speed control)
- Received Logic: Directional Logic (DIR)
- Logic Source: Jetson Orin NX GPIO
775 D-shaft Motors (x2)
Custom Coupling prevented interference
PWM (Speed)
DIR (Logic)
PWM (Logic)
RPWM
LPWM
REN
LEN
VCC
GND
B+
B-
M+
M-
Motor Output
Motor Output
Bidirectional Motor Drive
(to 755 D-shaft Motors)
- Executed Actions
- Coordinated Inward Rotation (for Ball Ingestion)
- Coordinated Outward Rotation (for Purging)
PWM (Speed)
PWM (Speed)
DIR (Logic)
Pugest: M+
Motor Output
Control Logic Isolation
(Optocoupler or Logic Isolator)
D-shaft
Ensures zero slippage